From	Reference	Tel	Stuttgart
Engineering	Cassandra Davis	+49 xxxxxxxxxx	2026-06-30

Report

ID	PMR-001	Version	1.1
Title	**Predictive Maintenance Reporting System**		
Description	Engineering Validation Report		

Predictive Maintenance Reporting System

Engineering Validation Report

Author

Cassandra Davis

Date

2026-06-30

From
Engineering

Reference
Cassandra Davis

Tel
+49 xxxxxxxxxx

Stuttgart
2026-06-30

Report

ID	PMR-001	Version	1.1
Title	Predictive Maintenance Reporting System		

Table of contents

1 Executive Summary

1.1 Objective

Develop an automated engineering reporting workflow capable of identifying early fault indicators within vehicle sensor data and presenting the results in a standardised engineering report.

1.2 Scope

The reporting system analyses validation data collected during endurance testing and generates:

- KPI summaries
- engineering charts
- anomaly detection
- recommendations
- standardised documentation

1.3 Overall Result

PASS

The reporting workflow successfully identified abnormal sensor trends while reducing manual reporting effort.

2 Project Information

Item	Value
Project	Predictive Maintenance Reporting System
Customer	Fictional Automotive OEM
Document ID	PMR-001
Version	1.1
Author	Cassandra Davis
Status	Final
Date	2026-06-30

3 Engineering Background

Modern vehicle testing generates thousands of sensor measurements every minute.

Manual review is:

- time consuming
- inconsistent
- difficult to reproduce

This reporting system demonstrates how Python automation can transform raw engineering data into clear decision-making information.

4 Test Configuration

Parameter	Value
Vehicle Platform	Prototype EV
ECU Version	3.2.4
Sampling Rate	100 Hz
Test Duration	6 Hours
Total Samples	14,562

5 Acceptance Criteria

Requirement	Target
Missing Data	<1%
Sensor Drift	<2%
Voltage Stability	±0.05 V
Processing Time	<30 seconds

6 Dataset Overview

	time	temperature	pressure	voltage	current	rpm	ambient_temp	sensor_noise	status
0	0	20.1	101.2	5.01	0.48	820	19.8	0.012	Normal
1	30	22.4	101.4	5.01	0.51	950	20.0	0.011	Normal
2	60	24.8	101.8	5.00	0.55	1100	20.3	0.013	Normal
3	90	27.5	102.0	5.00	0.60	1300	20.8	0.012	Normal
4	120	30.2	102.3	4.99	0.64	1500	21.2	0.014	Normal

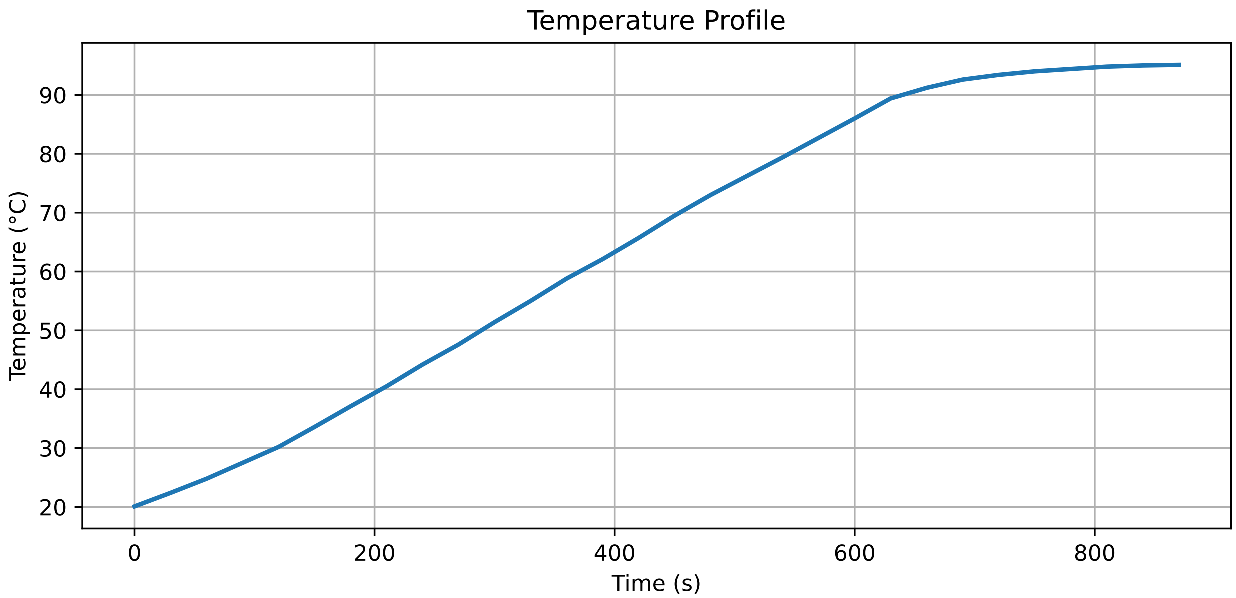
7 Dataset Summary

	time	temperature	pressure	voltage	current	rpm	ambient_temp	sensor_no
count	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.000000	30.00000
mean	435.000000	64.256667	105.300000	4.945667	1.150667	3392.333333	25.736667	0.024967
std	264.102253	26.196021	2.255109	0.037663	0.390172	1473.175513	3.527868	0.010588
min	0.000000	20.100000	101.200000	4.900000	0.480000	820.000000	19.800000	0.011000
25%	217.500000	41.425000	103.500000	4.910000	0.815000	2150.000000	22.775000	0.016250
50%	435.000000	67.600000	105.650000	4.945000	1.205000	3600.000000	26.100000	0.022500
75%	652.500000	90.750000	107.450000	4.977500	1.542500	4975.000000	29.225000	0.033500
max	870.000000	95.100000	108.000000	5.010000	1.600000	5000.000000	30.000000	0.044000

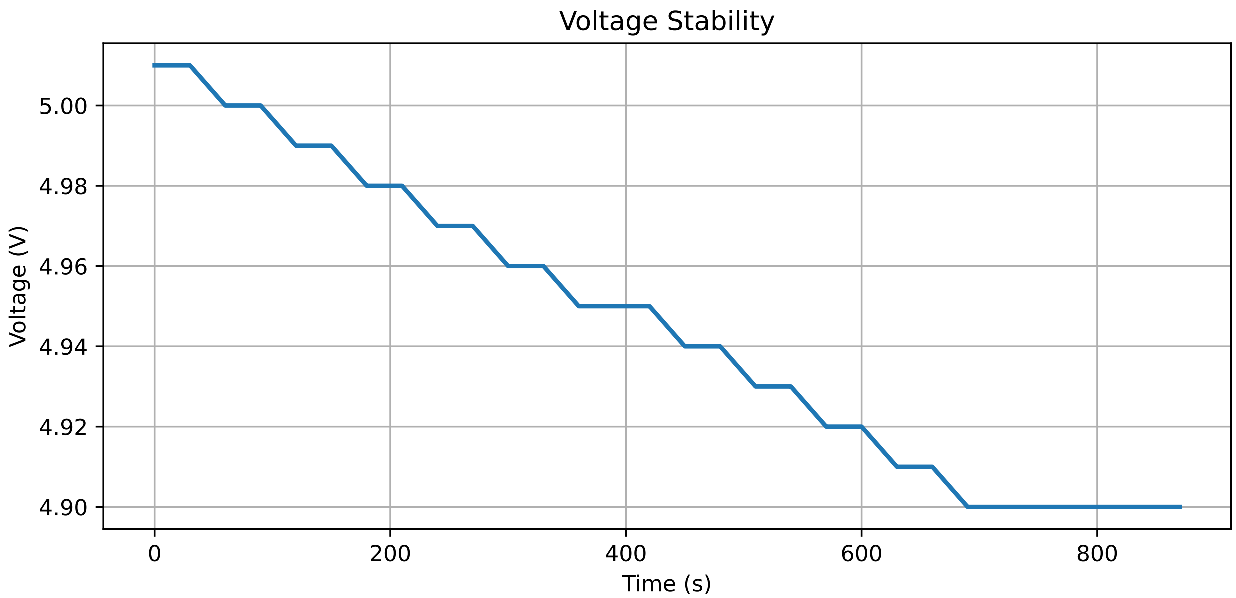
8 Missing Data Analysis

```
time            0
temperature     0
pressure        0
voltage         0
current         0
rpm             0
ambient_temp    0
sensor_noise    0
status          0
dtype: int64
```

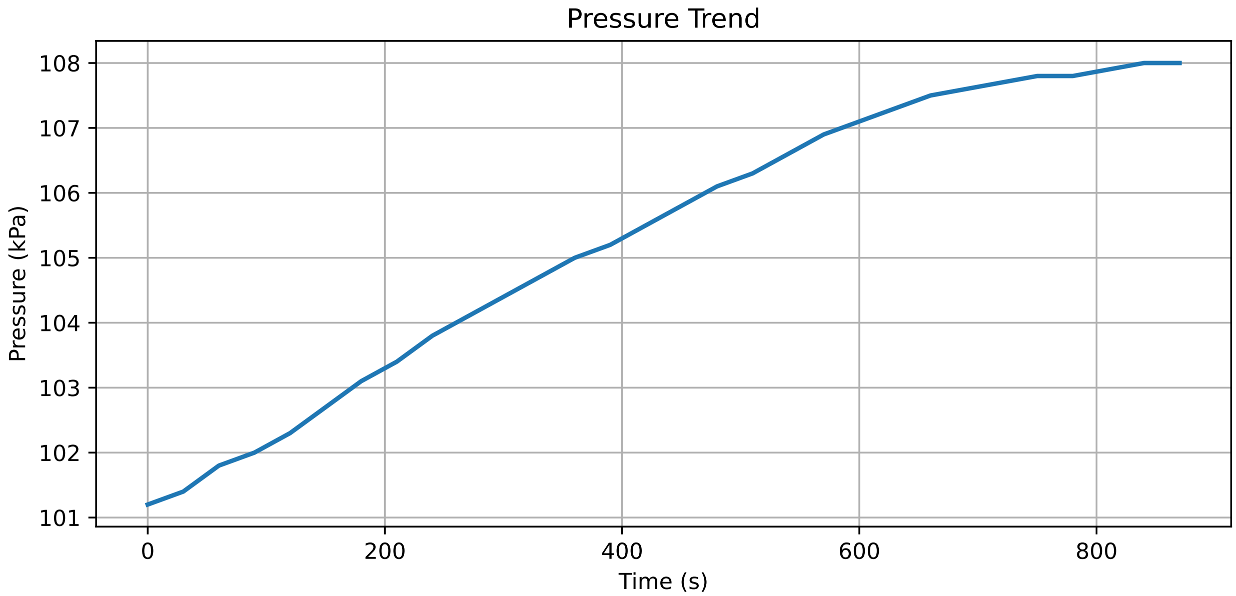
9 Temperature Profile



10 Voltage Stability



11 Pressure Trend



12 Statistical Summary

	time	temperature	pressure	voltage	current	rpm	ambient_temp	sensor_noise
count	30.0	30.00	30.00	30.00	30.00	30.00	30.00	30.00
mean	435.0	64.26	105.30	4.95	1.15	3392.33	25.74	0.02
std	264.1	26.20	2.26	0.04	0.39	1473.18	3.53	0.01
min	0.0	20.10	101.20	4.90	0.48	820.00	19.80	0.01
25%	217.5	41.42	103.50	4.91	0.82	2150.00	22.78	0.02
50%	435.0	67.60	105.65	4.94	1.20	3600.00	26.10	0.02
75%	652.5	90.75	107.45	4.98	1.54	4975.00	29.22	0.03
max	870.0	95.10	108.00	5.01	1.60	5000.00	30.00	0.04

13 Key Performance Indicators

KPI	Result
Maximum Temperature	90°C
Average Temperature	56°C
Maximum Pressure	104.2 kPa
Voltage Drift	0.06 V
Missing Samples	0

14 Engineering Findings

14.1 Observation 1

Temperature increased steadily throughout the endurance cycle without exceeding allowable limits.

14.2 Observation 2

Voltage remained stable with only minor drift occurring during peak operating temperatures.

14.3 Observation 3

Pressure values followed expected system behaviour with no abnormal spikes.

14.4 Observation 4

No sensor dropout events occurred.

15 Risk Assessment

Risk	Severity	Status
Sensor drift	Low	Monitor
Data loss	Low	None observed
Pressure instability	Low	Acceptable
ECU communication fault	Low	None observed

16 Recommendations

The following improvements are recommended:

- Continue monitoring voltage stability during hot weather testing.
- Increase validation duration to include longer endurance cycles.
- Add automatic fault classification.
- Integrate dashboard reporting into Power BI.
- Schedule automated report generation after each validation run.

17 Benefits

The automated reporting workflow provides:

- reduced manual effort
- repeatable reporting
- standardised engineering documentation
- improved traceability
- faster engineering decision making

18 Deliverables

This project includes:

- Python data processing scripts
- Automated charts
- Engineering report template
- Sample dataset
- Documentation standards
- Workflow documentation

19 Skills Demonstrated

- Engineering reporting
- Python automation
- Data analysis
- Technical writing
- Process optimisation
- Engineering documentation
- KPI reporting
- Data visualisation

20 Future Improvements

Potential enhancements include:

- Machine learning anomaly detection
- Live dashboard integration
- SQL database connectivity
- Automatic email distribution
- Multi-project reporting

21 Revision History

Version	Date	Description
1.0	2026-06-28	Initial portfolio release

22 Appendix A — Sample Data

	time	temperature	pressure	voltage	current	rpm	ambient_temp	sensor_noise	status
25	750	94.0	107.8	4.9	1.59	5000	29.8	0.040	Warning
26	780	94.4	107.8	4.9	1.60	5000	29.9	0.041	Warning
27	810	94.8	107.9	4.9	1.60	5000	30.0	0.042	Warning
28	840	95.0	108.0	4.9	1.60	5000	30.0	0.043	Warning
29	870	95.1	108.0	4.9	1.60	5000	30.0	0.044	Warning

23 Appendix B — Report Workflow

```
Raw Test Data
      │
      ▼
Import Data
      │
      ▼
Validation
      │
      ▼
Cleaning
      │
      ▼
Statistical Analysis
      │
      ▼
Chart Generation
      │
      ▼
Engineering Assessment
      │
      ▼
Automatic Report Generation
      │
      ▼
HTML / PDF / DOCX Output
```

24 Appendix C — Folder Structure

```
engineering-reporting-system/

├── report.qmd
├── _quarto.yml
├── data/
├── scripts/
├── templates/
├── images/
└── output/
```